Roll Number:

Name:

Section:

Indian Institute of Information Technology, Sri City, Chittoor

Spring Semester Schedule Quiz 2, April-2026

UG-1, Second Sem

Name of the Exam: P & S (S2) **Duration: 20 Minutes**

Apr 15, 2026 **Max. Marks: 10**

Instructions:

1. *Calculators are allowed.*
2. *All questions are mandatory and carry equal mark.*
3. *Tick only the right option. If we find two ticks for one question, '0' marks will be awarded.*
4. *Rough work should not be done in the question paper.*
5. *Return the question paper along with the rough sheet before leaving the exam hall.*

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1. **While tossing 100 coins, let H be the number of heads. What is the probability that $40 < H < 60$ using Chebyshev's inequality?**

   (a) 0.50

   (b) 0.75

   (c) 0.33

   (d) 0.67

   Answer:(b)

2. **Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with mean 5 and variance 9. As $n \to \infty$, $\dfrac{X_1^2 + X_2^2 + \cdots + X_n^2}{n} \xrightarrow{p} c$ for some constant c. Then the value of c is:**

   (a) 25

   (b) 34

   (c) 14

   (d) 28

   Answer:(b)

3. **If $E[X|Y] = 2Y + 1$, then $E[X]$ is**

   (a) $2E[Y]$

   (b) Cannot be determined

(P.T.O.)

(c) $2E[Y]+1$

(d) $E[Y]+2$

Answer: (c)

4. **Let** $X_1, \dots, X_7 \sim N(0, \sigma^2)$ **and** $Y_1, \dots, Y_9 \sim N(0, \sigma^2)$ **be two independent random samples. Define**

$$U = \sum_{i=1}^{7} X_i^2, \qquad V = \sum_{j=1}^{9} Y_j^2,$$

and consider the statistic

$$F = \frac{(U/7)}{(V/9)}.$$

Further, define the transformed variable

$$T = \frac{1}{F}.$$

Then T **follows an** F**-distribution with degrees of freedom** $(---,---)$**. (Enter your answer as an ordered pair)**

Answer:$(9, 7)$

5. **Let** X **and** Y **be independent random variables, each following the uniform distribution on** $(0, 1)$**. Which of the following best describes the probability density function of** $Z = X + Y$**?**

(a) Exponential distribution

(b) Triangular distribution

(c) Uniform distribution

(d) Normal distribution

Answer:(b)

6. ****Let** $f_{X,Y}(x, y) = xy,\ 0 < x < 2,\ 0 < y < 2$**. Which of the following is correct?**

(a) $E[XY] = E[X] + E[Y]$

(b) X and Y are independent

(c) $\mathrm{Cov}(X, Y) = 0$

(d) X and Y are not independent

Answer: (**)

7. **Let** $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$**. Consider the sample mean** $\bar{X}$ **and sample variance**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2,$$

and the maximum likelihood estimator (MLE) of variance

$$\hat{\sigma}^2_{\mathbf{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Which of the following statements are correct?

(a) $\bar{X}$ **is an unbiased estimator of** μ**, and** S^2 **is an unbiased estimator of** σ^2**.**

(b) $\bar{X}$ **is a consistent estimator of** μ**.**

(c) $\hat{\sigma}^2_{\mathbf{MLE}}$ **is biased but consistent for** σ^2**.**

(a) Only (a) is correct

(b) (a), (b), and (c) are all correct

(c) (a) and (b) are correct

(d) (b) and (c) are correct

Answer:(b)

8. **The average age of the cricket players is** 28.3 **years and has standard deviation of** 2.3 **years. If we can assume that ages are normally distributed, what is the probability that the average age of** 10 **randomly selected players is less than** 27 **years?**

   (a) $\Phi(-0.57)$

   (b) $\Phi(-1.79)$

   (c) $1 - \Phi(-1.79)$

   (d) $\Phi(0.57)$

   Answer:(b)

9. **Which of the following statements are correct?**

   (a) The estimator $T_n = \bar{X}_n + \frac{1}{n}$ **is biased and consistent.**

   (b) The estimator $T_n = X_1$ **is unbiased but not consistent.**

   (a) Both (a) and (b) are correct

   (b) Neither (a) nor (b) is correct

   (c) Only (a) is correct

   (d) Only (b) is correct

   Answer:(a)

10. **Let** $U \sim N(0, 1)$ **and** $V \sim \chi^2_\nu$ **be independent random variables. Define**

$$T = \frac{U}{\sqrt{V/\nu}}.$$

Let $\nu = 8$**. Consider the transformed variable**

$$W = \frac{T^2}{T^2 + \nu}.$$

Then W **follows a Beta distribution** $\mathrm{Beta}(a, b)$**. Find the values of** (a, b)**.**
(Enter your answer as an ordered pair $(\_\_\_, \_\_\_)$**)**

Answer: $\left(\frac{1}{2}, 4\right)$

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